

# Graph Neural Networks for Fraud Detection: A Survey and an Imbalance-Aware Approach

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**Abstract**—Fraud detection in online services can be formulated as semi-supervised node classification on graphs, where fraudsters are heavily outnumbered and actively camouflage themselves among benign users. This report surveys representative graph-based fraud detection methods — from the dense-block detector FRAUDAR to graph neural network (GNN) approaches such as SemiGNN, CARE-GNN, and PC-GNN — and outlines an imbalance-aware GNN approach to be implemented and evaluated on the YelpChi and Amazon fraud datasets. We adopt AUROC, AUPRC, and F1-macro as evaluation metrics, which are standard in this line of work.

**Index Terms**—graph neural networks, fraud detection, anomaly detection, class imbalance, semi-supervised learning

## I. INTRODUCTION

Online review and e-commerce platforms lose revenue and user trust to fraudulent accounts. Because fraudsters deliberately mimic benign behavior (*camouflage*) [1], purely feature-based detectors are easily evaded. Modeling users, reviews, and products as a graph and applying graph neural networks (GNNs) has therefore become a dominant paradigm; deep graph anomaly detection more broadly is surveyed in [2]. Two challenges are central: (1) extreme class imbalance between fraudsters and benign users, and (2) camouflage along both feature and relation dimensions. This project studies GNN-based fraud detectors that explicitly address these challenges.

## II. RELATED WORK

### A. Graph-based Fraud Detection without Deep Learning

A long line of work detects fraud through purely structural signals: groups of fraudulent accounts act in lockstep, creating unusually dense blocks in the adjacency matrix of, e.g., a user-product review graph. Density-based detectors exploit exactly this signal, but fraudsters evade them through *camouflage*: adding edges to honest, popular targets so as to “look normal.” FRAUDAR [1] is the representative method of this family. It detects dense subgraphs with a greedy, near-linear-time algorithm whose suspiciousness metric is designed to be robust to camouflage edges, and it provides provable, data-dependent bounds on the amount of fraud that can remain undetected on a given graph. Such methods are scalable and effective, but they ignore node features and the (scarce) available labels, which limits their accuracy on attributed graphs.

### B. Deep Graph Anomaly Detection

Ma et al. [2] systematize deep learning approaches to graph anomaly detection, organizing methods by the anomalous object they target (node, edge, subgraph, or graph level) and by architecture. Their survey highlights two challenges that are central to fraud detection and to this work: (1) extreme *class imbalance*, since anomalies are by definition rare, and (2) *camouflage*, since adversarial anomalies deliberately resemble normal instances. Fraud detection on review or transaction networks corresponds to node-level anomaly detection on attributed, multi-relation graphs under semi-supervision.

### C. GNN-based Fraud Detectors

Graph neural networks (GNNs) detect fraudulent nodes by aggregating neighborhood information. SemiGNN [3] is a semi-supervised graph attentive network developed for financial fraud detection: it organizes each node’s neighborhood into multiple views, applies hierarchical attention (node-level attention within a view, then view-level attention across views), and exploits unlabeled nodes through a graph-regularized semi-supervised loss. CARE-GNN [4] directly targets camouflage, distinguishing *feature camouflage* and *relation camouflage*. It introduces a label-aware similarity measure to identify informative neighbors, uses reinforcement learning to adapt a per-relation filtering threshold, and aggregates the filtered neighborhoods relation-wise. CARE-GNN also released the YelpChi and Amazon fraud benchmarks used in this study. Its main drawbacks are the cost and instability of the RL-based selector and that class imbalance is handled only implicitly.

### D. Imbalance-aware Graph Learning for Fraud Detection

Fraudsters typically constitute well under 15% of nodes, so naive GNN training is dominated by the majority class. PC-GNN [5] addresses this with a “pick and choose” strategy: a label-balanced sampler *picks* a balanced subset of labeled nodes for each training step, and a neighborhood sampler *chooses* neighbors by thresholding feature-space distances per relation, so that aggregation uses a balanced and informative neighborhood. A complementary, architecture-agnostic line of work handles imbalance at the loss level with class-weighted

TABLE I  
SURVEYED METHODS AND THE CHALLENGES THEY ADDRESS.

| Method       | Venue   | Camouflage | Imbalance |
|--------------|---------|------------|-----------|
| FRAUDAR [1]  | KDD'16  | yes        | no        |
| SemiGNN [3]  | ICDM'19 | no         | partial   |
| CARE-GNN [4] | CIKM'20 | yes        | no        |
| PC-GNN [5]   | WWW'21  | no         | yes       |
| <b>Ours</b>  | —       | yes        | yes       |

or focal objectives; we adopt a class-weighted cross-entropy in this work.

### E. Summary and Positioning

Table I contrasts the surveyed methods along the two challenges identified above. No single method addresses both camouflage and imbalance in a simple, stable way: FRAUDAR handles camouflage but ignores features and labels; CARE-GNN handles camouflage with an expensive RL selector; PC-GNN handles imbalance at the sampling level but ignores camouflage. Our approach (Section IV) combines a fixed, label-supervised similarity filter in the spirit of CARE-GNN — removing the RL machinery — with PC-GNN-style balanced neighborhood sampling and a class-weighted objective, and we ablate each component to measure its contribution.

## III. PROBLEM DEFINITION

Let  $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$  be an attributed graph with node set  $\mathcal{V}$ , edge set  $\mathcal{E}$ , and node feature matrix  $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$ . Each node  $v \in \mathcal{V}$  has a label  $y_v \in \{0, 1\}$  ( $1 = \text{fraudster}$ ), of which only a small labeled subset  $\mathcal{V}_L \subset \mathcal{V}$  is observed, with  $|\{v : y_v = 1\}| \ll |\{v : y_v = 0\}|$  (class imbalance). The task is *semi-supervised node classification*: learn  $f : \mathcal{V} \rightarrow [0, 1]$  from  $G$  and  $\mathcal{V}_L$  that ranks fraudulent nodes above benign ones on the unlabeled nodes.

## IV. PROPOSED METHOD

TODO: We plan to implement an imbalance-aware GNN combining (i) camouflage-resistant neighbor selection in the spirit of CARE-GNN [4] and (ii) balanced neighborhood sampling as in PC-GNN [5], trained with a semi-supervised objective following SemiGNN [3].

## V. EXPERIMENTS

### A. Datasets

We use two public benchmarks: **YelpChi** (Yelp review spam, relations: R-U-R, R-S-R, R-T-R) and **Amazon** (Amazon product review fraud, relations: U-P-U, U-S-U, U-V-V), both released with CARE-GNN [4].

### B. Baselines

GCN, SemiGNN [3], CARE-GNN [4], and PC-GNN [5].

### C. Metrics

AUROC, AUPRC, and F1-macro, the standard metrics for this benchmark [4], [5].

## VI. CONCLUSION AND FUTURE WORK

TODO: summarize findings after experiments; future work includes exploring heterophily-aware GNNs and cost-sensitive learning for fraud detection.

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